

# RobotArm\_Research\_Plan\_Fall2026\_v2

## Fall 2026 Robot Arm Research Plan

**Embodied AI demos for Spring 2027 and a viewpoint robustness study with a digital twin**

Team: Aakash and Ayush | Faculty lead: Dr. Guannan Liu | 9/14/2026 to 12/18/2026

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### 1. What we will have by December

1. **A Spring 2027 demo package.** Imitation learning (ACT), VLA (SmolVLA), and reinforcement learning, each runnable from a README, plus an ACT demo on the large arm. AI agent control is a stretch demo.
2. **One conference paper.** Target: IEEE BMSB (deadline to be confirmed, around mid December).
3. **A sim and real evaluation record** on the same task, which becomes course material in the spring.

### 2. Where we start

Everything is installed and working: SO101, the two large arms, Isaac Sim, and the ACT and SmolVLA training pipelines. Last round, both policies struggled because the camera position changed between data collection and testing. So the first job is simple: fix the camera, recollect the data, retrain, and move on quickly.

That camera problem is also our research question. Both policies broke when the viewpoint moved. We will measure exactly how much, and whether a digital twin can predict it.

### 3. How we work

- **Both of you do everything.** Real arm, simulation, large arm, writing. Tasks are split evenly, and owners rotate so each of you runs every pipeline at least once.
- **Every task has an owner.** In the weekly tables, the owner is the person who makes sure it is done. "Both" means you do it together, usually because the task needs two people (all large arm sessions do).
- **Deliverables go in the shared repository by Friday 11:59 pm**, in the exact path listed. A deliverable is done when it meets its "Done when" line.
- **One weekly report every Friday** (template in Section 9). If something blocks you for two work sessions, message Dr. Liu the same day.

- **Gate reviews** with Dr. Liu happen in person after Weeks 2, 6, and 9. Passing a gate unlocks the next level on the large arm.

## 4. Research questions

- **RQ1 (core).** How sensitive are ACT and SmoIvLA to camera viewpoint shift on a low cost arm? Does SmoIvLA's pretraining make it more robust than ACT?
- **RQ2 (core).** Does an Isaac Sim digital twin of our setup show the same drop in performance under the same viewpoint shifts? If yes, a cheap sweep in simulation can predict where the real policy will fail.
- **RQ3 (stretch).** Does training on data from several viewpoints recover performance at a new viewpoint?

Note on RQ2: we compare the **trend** (how much success falls as the camera moves) in sim and in real, not the absolute success rate. A policy trained on real images will not run well on rendered images, and we do not need it to.

## 5. Repository layout

```
docs/
  eval_protocol.md
  camera_poses.md
  related_work.md
  runbook_act.md
  runbook_smolvla.md
  runbook_bigarm.md
  img/
results/
  results.csv          # one row per trial, real and sim
videos/
  W02/ W03/ ...       # one folder per week
sim/
  scene_poseA.usd
  expert_policy.py
bigarm/
  hello.py
  scripted_pick.py
analysis/
  plot_viewpoint.py
  make_all_figures.py
figures/
reports/
  W01.md ... W14.md
  midpoint.md
```

```
paper/  
demos/  
  act/ smolvla/ rl/ bigarm_act/ agent/
```

**results.csv columns:** date, domain (real/sim), robot (so101/bigarm), policy, checkpoint, train\_data, camera\_pose, init\_pos\_id, trial, success (0/1), failure\_mode, operator, video\_file, notes

**Video naming:** videos/W02/act\_poseA\_pos03\_t1.mp4 (policy, pose, initial position, trial).

**Hugging Face datasets:** <hf\_org>/so101\_pick\_poseA\_v2 , <hf\_org>/so101\_pick\_sim\_v1 , and so on. Every dataset gets a card listing episodes, cameras, pose, and collector.

## 6. Evaluation protocol

Written in Week 1 and frozen before any new data is collected. Every number in the paper comes from this protocol.

- **Task:** the current pick and place task, stated in one sentence with a start region and a goal region.
- **Demonstrations:** 50 per training condition.
- **Initial positions:** 10 positions marked on the table with tape and numbered 1 to 10, photographed from above.
- **Trials:** each initial position tested twice, so **20 trials per condition**.
- **Success:** the object is released inside the goal region within the time limit written in the protocol. A trial the operator has to stop for safety counts as a failure.
- **Checkpoint rule:** the checkpoint is chosen before evaluation (fixed step or last checkpoint). Never chosen by looking at test results.
- **Camera poses:** rigid mount, base outlined with tape. Pose A is the training pose. Each pose is recorded as position from the arm base (x, y, z in cm) and angle (yaw, pitch in degrees), plus a photo.
- **Start of every session:** take one photo from the camera and compare it with the pose A reference photo. If they do not match, fix the mount before collecting or testing anything.
- **ACT threshold for Gate 1:** set by Dr. Liu and written into the protocol before the first evaluation.

## 7. Large arm: levels

Access is earned by finishing work, not by hours. Everyone who passes a gate moves up.

**Safety first (Week 2, required before any level).** Training session with Dr. Liu, and a signed lab SOP:

- Emergency stop within reach at all times.
- Reduced speed and collision detection on for every session.
- Workspace marked on the floor; nobody inside it while the arm moves on its own.
- Two people present whenever the arm is powered. That is why every large arm task below is owned by "Both".

| Level   | Unlocked by            | What you get to do                                                               |
|---------|------------------------|----------------------------------------------------------------------------------|
| Level 1 | Gate 1 (end of Week 2) | Jog the arm, run scripted motions, scripted pick and place with the SDK          |
| Level 2 | Gate 2 (end of Week 6) | Teleoperate, collect your own demonstrations, train and run ACT on the large arm |
| Level 3 | Gate 3 (end of Week 9) | Open exploration time on your own ideas, within the SOP                          |

## 8. Weekly plan

Owner key: **Aakash**, **Ayush**, **Both**. Stretch tasks are marked (*stretch*).

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### Week 1 (9/14 to 9/18): Fix the setup, recollect the data

**Goal:** a correct pose A and 50 clean demonstrations on the Hub.

| #   | Task                                                                        | Owner  |
|-----|-----------------------------------------------------------------------------|--------|
| 1.1 | Mount every training camera rigidly; outline the mount bases with tape      | Aakash |
| 1.2 | Measure pose A (x, y, z, yaw, pitch) and take the reference photos          | Ayush  |
| 1.3 | Mark and photograph initial positions 1 to 10                               | Ayush  |
| 1.4 | Write docs/eval_protocol.md using Section 6                                 | Aakash |
| 1.5 | Review the protocol and flag anything unclear                               | Ayush  |
| 1.6 | Collect 25 demonstrations at positions 1 to 5 (5 each)                      | Aakash |
| 1.7 | Collect 25 demonstrations at positions 6 to 10 (5 each)                     | Ayush  |
| 1.8 | Replay 5 random episodes to check them; push the dataset and write its card | Aakash |
| 1.9 | Start ACT training on the new dataset                                       | Ayush  |

### Deliverables

- docs/eval\_protocol.md . Done when: all items in Section 6 are filled in except the Gate 1 threshold.
- docs/camera\_poses.md + docs/img/poseA\*.jpg + docs/img/init\_positions.jpg . Done when: pose A has numbers and photos, and all 10 positions are visible and numbered in one photo.
- HF dataset so101\_pick\_poseA\_v2 . Done when: 50 episodes, card lists collector per episode range, 5 replay checks passed.
- reports/W01.md .

## Week 2 (9/21 to 9/25): ACT and SmolVLA working again

**Goal:** both policies evaluated at pose A, and each of you has run both pipelines.

| #   | Task                                                                                | Owner  |
|-----|-------------------------------------------------------------------------------------|--------|
| 2.1 | Finish ACT training; run 10 trials (positions 1 to 5, twice each)                   | Ayush  |
| 2.2 | Fine tune SmolVLA on the new dataset; run 10 trials (positions 6 to 10, twice each) | Aakash |
| 2.3 | Run the other 10 ACT trials (positions 6 to 10)                                     | Aakash |
| 2.4 | Run the other 10 SmolVLA trials (positions 1 to 5)                                  | Ayush  |
| 2.5 | Write docs/runbook_act.md : every command from raw data to evaluation               | Ayush  |
| 2.6 | Write docs/runbook_smolvla.md : every command from raw data to evaluation           | Aakash |
| 2.7 | Large arm safety training and SOP signature                                         | Both   |

### Deliverables

- results/results.csv with 40 rows (20 ACT, 20 SmolVLA, pose A). Done when: every row has a matching video in videos/W02/ .
- Checkpoints for both policies on the Hub, with the training step recorded in the protocol.
- Two runbooks. Done when: the other person can follow the runbook without asking a question (Task 2.3 and 2.4 are the test).
- Signed SOP in docs/ .
- reports/W02.md with the pose A success table.

**Gate 1 review.** Pass when: ACT meets the protocol threshold at pose A, SmolVLA is evaluated, both runbooks work, safety training done. **Unlocks Level 1.**

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## Week 3 (9/28 to 10/2): First time on the large arm, first viewpoint test

**Goal:** the first real viewpoint shift results, and the large arm moving under your code.

| #   | Task                                                                                   | Owner  |
|-----|----------------------------------------------------------------------------------------|--------|
| 3.1 | Define poses B (lateral shift) and C (yaw rotation); measure and photograph them       | Aakash |
| 3.2 | Evaluate ACT at poses B and C, 20 trials each                                          | Aakash |
| 3.3 | Evaluate SmolVLA at poses B and C, 20 trials each                                      | Ayush  |
| 3.4 | Large arm session 1: power up, e-stop test, jog, move through 5 named poses            | Both   |
| 3.5 | Write <code>bigarm/hello.py</code> (connect, reduced speed, move to named poses, home) | Ayush  |
| 3.6 | Start <code>docs/runbook_bigarm.md</code>                                              | Aakash |

### Deliverables

- `docs/camera_poses.md` updated with B and C.
- 80 new rows in `results.csv`, videos in `videos/W03/`.
- `figures/viewpoint_real_v1.png`: success rate for 2 policies across poses A, B, C.
- `bigarm/hello.py` + `videos/W03/bigarm_hello.mp4`.
- `reports/W03.md`.

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## Week 4 (10/5 to 10/9): Build the digital twin

**Goal:** an Isaac Sim scene that looks like our real setup from pose A.

| #   | Task                                                                                                      | Owner  |
|-----|-----------------------------------------------------------------------------------------------------------|--------|
| 4.1 | Import SO101 into Isaac Sim; check joint limits against the real arm                                      | Ayush  |
| 4.2 | Build the table, object, and goal region to measured dimensions                                           | Aakash |
| 4.3 | Place the sim camera at pose A; render and compare with the real reference photo; adjust until they match | Both   |
| 4.4 | Large arm session 2: scripted pick and place with the SDK                                                 | Both   |
| 4.5 | Write the pick half of <code>bigarm/scripted_pick.py</code>                                               | Aakash |

| #   | Task                                                                                                                                      | Owner |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4.6 | Write the place and home half of <code>bigarm/scripted_pick.py</code>                                                                     | Ayush |
| 4.7 | Literature scan: 5 papers each on viewpoint robustness and evaluating real robot policies in simulation (start from SIMPLER and RoVi-Aug) | Both  |

## Deliverables

- `sim/scene_poseA.usd` .
- `docs/img/sim_vs_real_poseA.jpg` : real and sim frames side by side. Done when: arm base, table edge, and goal region line up by eye.
- `bigarm/scripted_pick.py` + `videos/W04/bigarm_scripted_pick.mp4` .
- `docs/related_work.md` : 10 rows (citation, what they did, how our work differs), 5 per person, name on each row.
- `reports/W04.md` .

## Week 5 (10/12 to 10/16): Simulated data and ACT in simulation

**Goal:** ACT trained in sim and tested under the same viewpoint shifts as real.

| #   | Task                                                                                                | Owner  |
|-----|-----------------------------------------------------------------------------------------------------|--------|
| 5.1 | Write the scripted expert in sim ( <code>sim/expert_policy.py</code> )                              | Ayush  |
| 5.2 | Set object starts to the same 10 initial positions; export episodes in LeRobot format               | Aakash |
| 5.3 | Generate 50 sim episodes and push them to the Hub                                                   | Ayush  |
| 5.4 | Train ACT on sim data                                                                               | Aakash |
| 5.5 | Evaluate sim ACT at sim poses A, B, C, 20 trials each, logged to <code>results.csv</code>           | Ayush  |
| 5.6 | Real generalization check: 3 new initial positions and 1 distractor object, ACT and SmoIVLA, pose A | Aakash |

## Deliverables

- HF dataset `so101_pick_sim_v1` with card.
- 60 sim rows and the generalization rows in `results.csv` .
- `figures/viewpoint_sim_v1.png` .
- `reports/W05.md` .

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## Week 6 (10/19 to 10/23): Sim versus real, and the midpoint report

**Goal:** answer the first version of RQ1 and RQ2.

| #   | Task                                                                                           | Owner  |
|-----|------------------------------------------------------------------------------------------------|--------|
| 6.1 | Dense sim sweep: 7 camera poses, 20 trials each                                                | Ayush  |
| 6.2 | Write <code>analysis/plot_viewpoint.py</code> : real and sim success versus pose on one figure | Aakash |
| 6.3 | Pick one new real pose D where sim predicts a large drop; run ACT there, 20 trials             | Aakash |
| 6.4 | Run SmolVLA at pose D, 20 trials                                                               | Ayush  |
| 6.5 | Midpoint report, sections 1 and 2 (setup, real results)                                        | Aakash |
| 6.6 | Midpoint report, sections 3 and 4 (sim results, did sim predict pose D)                        | Ayush  |

### Deliverables

- `reports/midpoint.md` : 2 pages, 3 figures, one paragraph on what surprised you.
- `figures/sim_vs_real.png` .
- `results.csv` complete for poses A, B, C, D (real) and the 7 pose sweep (sim).

**Gate 2 review.** Pass when: the midpoint report is complete and every number traces to rows in `results.csv` . **Unlocks Level 2.**

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## Week 7 (10/26 to 10/30): Level 2 begins, multi view data

**Goal:** first demonstrations on the large arm, and the robustness experiment started.

| #   | Task                                                                                     | Owner  |
|-----|------------------------------------------------------------------------------------------|--------|
| 7.1 | Large arm: set up teleoperation and recording in LeRobot format                          | Both   |
| 7.2 | Record 10 test demonstrations on the large arm and check replay                          | Both   |
| 7.3 | Collect 25 SO101 demonstrations at pose B ( <i>stretch</i> , RQ3)                        | Aakash |
| 7.4 | Collect 25 SO101 demonstrations at pose C ( <i>stretch</i> , RQ3)                        | Ayush  |
| 7.5 | Sim multi view: generate episodes at poses A, B, C; train sim ACT; test across the sweep | Ayush  |
| 7.6 | Update <code>docs/runbook_bigarm.md</code> with the teleoperation steps                  | Aakash |

## Deliverables

- 10 large arm episodes on the Hub ( `bigarm_pick_v0` ) + `videos/W07/bigarm_teleop.mp4` .
  - Real multi view dataset (*stretch*).
  - Sim multi view rows in `results.csv` .
  - `reports/W07.md` .
- 

## Week 8 (11/2 to 11/6): Reinforcement learning demo, large arm data

**Goal:** an RL demo in simulation and a full large arm dataset.

| #   | Task                                                                                   | Owner  |
|-----|----------------------------------------------------------------------------------------|--------|
| 8.1 | RL reach task for SO101 with PPO in Isaac Lab (MuJoCo if Isaac Lab is not ready)       | Ayush  |
| 8.2 | Try the trained reach policy on the real SO101 (joint targets only) ( <i>stretch</i> ) | Aakash |
| 8.3 | Large arm: 40 more demonstrations, 20 each, 50 total                                   | Both   |
| 8.4 | Train ACT on the large arm dataset                                                     | Aakash |
| 8.5 | Train real multi view ACT; test at pose D, 20 trials ( <i>stretch</i> )                | Ayush  |

## Deliverables

- `figures/rl_reward_curve.png` + `videos/W08/rl_reach_sim.mp4` (+ real attempt video if 8.2 done).
  - HF dataset `bigarm_pick_v1` (50 episodes).
  - Large arm ACT checkpoint.
  - `reports/W08.md` .
- 

## Week 9 (11/9 to 11/13): Freeze the experiments

**Goal:** every experiment for the paper finished. Veterans Day is 11/11.

| #   | Task                                                                           | Owner |
|-----|--------------------------------------------------------------------------------|-------|
| 9.1 | Large arm ACT evaluation, 20 trials (operator and safety spotter swap halfway) | Both  |

| #   | Task                                                                                                          | Owner  |
|-----|---------------------------------------------------------------------------------------------------------------|--------|
| 9.2 | Find missing cells in <code>results.csv</code> ; rerun any condition with fewer than 20 trials                | Aakash |
| 9.3 | Write <code>analysis/make_all_figures.py</code> : every paper figure from <code>results.csv</code> in one run | Ayush  |
| 9.4 | Paper outline in <code>paper/outline.md</code> with one line per paragraph                                    | Aakash |
| 9.5 | Failure mode breakdown: label every failed trial into 3 to 5 categories                                       | Ayush  |

## Deliverables

- Frozen `results.csv` (tagged `results-v1` in git).
- `figures/` regenerated by one script.
- `paper/outline.md` .
- `videos/W09/bigarm_act_eval/` .
- `reports/W09.md` .

**Gate 3 review.** Pass when: results are frozen, figures regenerate from one command, outline agreed with Dr. Liu. **Unlocks Level 3.**

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## Week 10 (11/16 to 11/20): Paper draft v1

| #    | Task                                                                                                                               | Owner   |
|------|------------------------------------------------------------------------------------------------------------------------------------|---------|
| 10.1 | Setup section: task, platforms, protocol                                                                                           | Aakash  |
| 10.2 | Related Work section from <code>docs/related_work.md</code>                                                                        | Ayush   |
| 10.3 | Results and Discussion for RQ1                                                                                                     | Aakash  |
| 10.4 | Results and Discussion for RQ2 (and RQ3 if done)                                                                                   | Ayush   |
| 10.5 | Introduction and Abstract                                                                                                          | Dr. Liu |
| 10.6 | AI agent demo: skill API on SO101 ( <code>pick(pos)</code> , <code>place(pos)</code> , <code>home()</code> )<br>( <i>stretch</i> ) | Ayush   |

## Deliverables

- Full draft v1 in `paper/` (or the Overleaf link in the report), sent to Dr. Liu Friday.
  - `reports/W10.md` .
-

## Week 11 (11/23 to 11/27): Demo packages (Thanksgiving week, lighter load)

| #    | Task                                                                             | Owner  |
|------|----------------------------------------------------------------------------------|--------|
| 11.1 | Turn <code>runbook_act.md</code> into <code>demos/act/</code> (Section 10 spec)  | Ayush  |
| 11.2 | Turn <code>runbook_smolvla.md</code> into <code>demos/smolvla/</code>            | Aakash |
| 11.3 | Record a one minute video for each                                               | Both   |
| 11.4 | AI agent demo: LLM tool calling loop that calls the skill API ( <i>stretch</i> ) | Aakash |

**Deliverables:** `demos/act/` and `demos/smolvla/` complete, `reports/W11.md` .

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## Week 12 (11/30 to 12/4): Paper v2 and the remaining demos

| #    | Task                                               | Owner  |
|------|----------------------------------------------------|--------|
| 12.1 | Revise your paper sections from Dr. Liu's comments | Both   |
| 12.2 | <code>demos/rl/</code>                             | Ayush  |
| 12.3 | <code>demos/bigarm_act/</code>                     | Aakash |
| 12.4 | <code>demos/agent/</code> ( <i>stretch</i> )       | Both   |

**Deliverables:** paper v2, all demo folders, `reports/W12.md` .

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## Week 13 (12/7 to 12/11): Cross check (finals week, lighter load)

| #    | Task                                                                                                      | Owner  |
|------|-----------------------------------------------------------------------------------------------------------|--------|
| 13.1 | Reproduce <code>demos/act/</code> and <code>demos/rl/</code> from scratch on a clean machine; file issues | Aakash |
| 13.2 | Reproduce <code>demos/smolvla/</code> and <code>demos/bigarm_act/</code> from scratch; file issues        | Ayush  |
| 13.3 | Paper v3: figures, references, page limit, format                                                         | Both   |

**Deliverables:** every demo reproduced by the person who did not build it, paper v3 submission ready, `reports/W13.md` .

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## Week 14 (12/14 to 12/18): Submit and hand off

| #    | Task                                                                        | Owner  |
|------|-----------------------------------------------------------------------------|--------|
| 14.1 | Submit the paper; save the confirmation                                     | Aakash |
| 14.2 | Tag the repository v1.0 ; check every dataset and checkpoint link           | Ayush  |
| 14.3 | Final report: what works, what does not, what the spring course should know | Both   |

**Deliverables:** submission confirmation in `paper/` , repository tag `v1.0` , `reports/final.md` .

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## 9. Weekly report template

`reports/WXX.md` , due Friday 11:59 pm. One report, one section per person.

```
# Week XX report

## Aakash
- Tasks done (by number):
- Hours this week, by task:
- Links to deliverables:
- Numbers (success rates, episodes, training steps):
- Blocker and what I tried:

## Ayush
(same five lines)

## Next week
- Plan by task number and owner:
- Current large arm level:
```

## 10. Spring demo package spec

Each folder in `demos/` contains:

- `README.md` with one command to run the demo, and the hardware and cameras needed.
- An environment file.
- Links to the dataset and the checkpoint on the Hub.
- A one minute video of the demo working.

- Known failure modes, with one video of a typical failure.
- `teaching_note.md` , one page: what students should notice, and why it happens.

## 11. Paper plan

**Venue:** IEEE BMSB (2027 call to be confirmed). **Authors:** Aakash, Ayush, Guannan Liu (corresponding).

### Outline

1. Introduction
2. Related Work
3. Setup: task, platforms, evaluation protocol
4. Viewpoint sensitivity of ACT and SmoIVLA (RQ1)
5. Does the digital twin predict the drop? (RQ2)
6. Multi view training (*if done*)
7. Discussion: what this means for low cost robot learning and for teaching
8. Conclusion

### Figures

- F1: setup photo with camera poses A to D marked
- F2: success rate versus camera pose, ACT and SmoIVLA
- F3: sim and real success curves on the same axes
- F4: failure mode breakdown
- T1: protocol summary